Similarly, $\lim_{x\to -3^-} \frac{|x|}{x} = \lim_{x\to -3^-} \frac{-x}{x} = -1$

$$\lim_{x\to -3^+} \frac{|x|}{x} = \lim_{x\to -3^+} \frac{-x}{x} = -1.$$

In fact, for any real number $x_0 \neq 0$, $\lim_{x\to x_0^-} \frac{|x|}{x} = -1 = \lim_{x\to x_0^+} \frac{|x|}{x}$ if $x_0 < 0$ and

$$\lim_{x\to x_0^-} \frac{|x|}{x} = 1 = \lim_{x\to x_0^+} \frac{|x|}{x} \text{ if } x_0 > 0.$$

We call the attention of the reader to observe the differences reflected in Illustrations 9.1 to 9.3. In Illustration 9.1, the function $f(x) = x^2 + 3$ is defined at $x = 2$. i.e., 2 belongs to the domain of f namely $\mathbb{R} = (-\infty, \infty)$. In Illustration 9.2, the function is not defined at $x = -4$. In the former case we say the limit, $\lim_{x\to 2} f(x)$ exists as x gets closer and closer to 2 to mean that $\lim_{x\to 2^-} f(x)$ and $\lim_{x\to 2^+} f(x)$ stand for a unique real number. In the later case, although it is not defined at $x = -4$, $\lim_{x\to -4} f(x)$ exist as x gets closer and closer to -4. In Illustration 9.3, $\lim_{x\to 0} \frac{|x|}{x}$ does not exist to mean that the one sided limits $\lim_{x\to 0^-} \frac{|x|}{x}$ and $\lim_{x\to 0^+} \frac{|x|}{x}$ are different as x gets sufficiently close to 0. In the light of these observations we have the **intuitive** notion of limit as in

Definition 9.1

Let I be an open interval containing $x_0 \in \mathbb{R}$. Let $f : I \to \mathbb{R}$. Then we say that the limit of $f(x)$ is L, as x approaches x_0 [symbollically written as $\lim_{x\to x_0} f(x) = L$], if, whenever x becomes sufficiently close to x_0 from either side with $x \neq x_0$, $f(x)$ gets sufficiently close to L.

The following (Fig 9.4 and 9.5) graphs depict the above narrations.

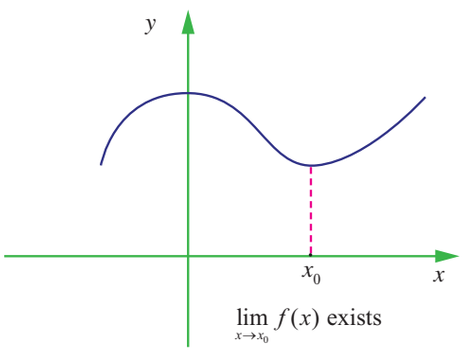


Fig. 9.4

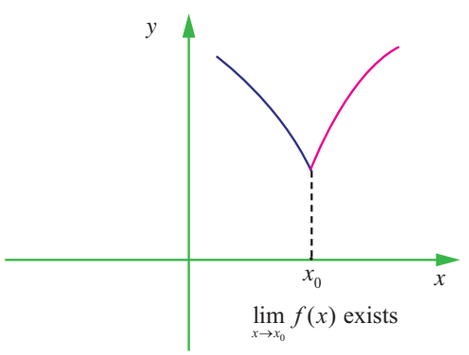


Fig. 9.5

9.2.2 One sided limits

Definition 9.2

We say that the **left-hand limit** of $f(x)$ as x approaches x_0 (or the limit of $f(x)$ as x approaches x_0 from the left) is equal to l_1 if we can make the values of $f(x)$ arbitrarily close to l_1 by taking x to be sufficiently close to x_0 and less than x_0. It is symbolically written as $f(x_0^-) = \lim_{x\to x_0^-} f(x) = l_1$.

Similarly, we define the right hand limit.

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